ISMAEL ORTIZ

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EDUCATION

RICE UNIVERSITY — **Houston, TX**
Ph.D. in Bioengineering — dissertation defense November 2026 — *2024 – 2026*

- Doctoral work begun at **Vanderbilt University** (2020–2024); relocated with research group to Rice
- Thesis: *Cancer Cell Migration and Adaptation to Fluid Shear Stress: Transcriptional, Phenotypic, and Metabolic Determinants*
- Awarded two national predoctoral fellowships (NSF Graduate Research Fellowship; Ford Foundation Predoctoral Fellowship), an NIH Diversity Supplement, Provost's Fellowships at both institutions, and the Compton Prize for the best first-year science and engineering research proposal at Vanderbilt
- 8 peer-reviewed publications (4 first or co-first author) including *PNAS*, *Biomaterials*, and *EMBO Reports*; 14 conference presentations

UNIVERSITY OF MIAMI — **Coral Gables, FL**
B.S. in Biomedical Engineering, Pre-Med Concentration — *2016 – 2020*

- Dean's List; Canes Achievement Award; Florida Bright Futures Medallion Scholar

RESEARCH & PROFESSIONAL EXPERIENCE

REINHART-KING LABORATORY — Rice and Vanderbilt Universities — **Houston, TX**
Graduate Researcher — *2020 – Present*

- Set the direction of a fellowship-funded, six-year research program on cancer cell metabolism and tumor cell survival, framing the question, work plan, and analytical approach independently.
- **Cancer cell metabolism:** quantified metabolic reprogramming in tumor cells by Agilent Seahorse extracellular flux analysis, isolating a calcium-channel-dependent mechanism that anchors a manuscript in preparation. Publications: 10.1038/s44319-026-00795-4
- **Transcriptomics:** structured ambiguous, high-volume sequencing data across multiple cancer types into a defensible co-first-author conclusion on the shared regulatory programs driving cancer cell migration; built reproducible R and MATLAB pipelines adopted across the group. Publications: https://doi.org/10.3390/genes17091042 , https://doi.org/10.1242/jcs.265092
- **Tumor immunology:** characterized how engineered liposome delivery alters T cell migration through tissue, published as co-first author in a study directed at cancer immunotherapy design. Publications: 10.1016/j.jim.2024.113700
- Recruited and directed a mentee team spanning a masters student, three undergraduates, and a high school student — coaching the high school mentee to first authorship on a peer-reviewed paper and undergraduates to co-authorship on my first-author work.
- Drove a three-lab, multi-institution collaboration to a completed manuscript and communicated findings across 14 conference presentations and an invited review synthesizing an emerging field for a broad readership.

PHYSIOMIMETIC MICROSYSTEMS LABORATORY — University of Miami — **Miami, FL**
Undergraduate Researcher — *2018 – 2020*

- Designed a two-layer vascular wall model of circulating tumor cell metastasis on-chip; presented at the national BMES conference and co-authored a resulting publication.

WATER QUALITY ENGINEERING LABORATORY — University of Miami — **Miami, FL**
Undergraduate Researcher — *2018 – 2019*

- Brokered a cross-university partnership with Florida International University and co-developed a flow reactor recovering nitrogen and phosphorus from municipal wastewater, targeting a lower unit cost of fertilizer production.

LEADERSHIP & MENTORSHIP

SyBBURE SEARLE GRADUATE FELLOW — Vanderbilt University — **Nashville, TN**
Undergraduate Research Program Mentor — *2023 – 2024*

- Selected as one of a small cohort of fellows guiding a ~50-student undergraduate research program; led a 4–5 student team weekly through project scoping, road mapping, and delivery of minimum viable prototypes.

ADDITIONAL

Expertise: Cancer cell metabolism, tumor immunology, transcriptomics, mechanobiology, molecular and cell biology
Analytical: R, MATLAB, Excel, RNA-sequencing analysis, Agilent Seahorse XF, FlowJo, ImageJ, SolidWorks
Languages: Native Spanish; fluent English
Interests: Documentary filmmaking — directed and edited "Independence with Autism" for South Florida PBS; https://youtu.be/nCU8RRaSiAA?si=mpMjxapp1KOzahaV
Served as Sebastian the Ibis, the University of Miami mascot; https://miamihurricanes.com/news/2020/04/07/being-sebastian
Camp Kesem counselor for children whose parents have cancer